Lecture #10 | Exploratory data analysis (cont.)

SE377 Introduction to Big Data Analysis and Visualization (2017)

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Announcement

- Project team is announced
  - Say hi to your teammates (in case you haven’t done so)
  - Start working on term project
    - Select a topic
    - Search if you can get relevant data

A sample case

- Would you invest in this company?

	January	February	March	April	May
Rev/customer	$5.00	$4.50	$4.33	$4.25	$4.50

Source: Lean Analytics

A sample case (cont.)

- Revenue per customer by cohort

Cohort	January	February	March	April	May
January	$5	$3	$2	$1	$0.5
February		$6	$4	$2	$1
March			$7	$6	$5
April				$8	$7
May					$9
Average	**$5.00**	**$4.50**	**$4.33**	**$4.25**	**$4.50**

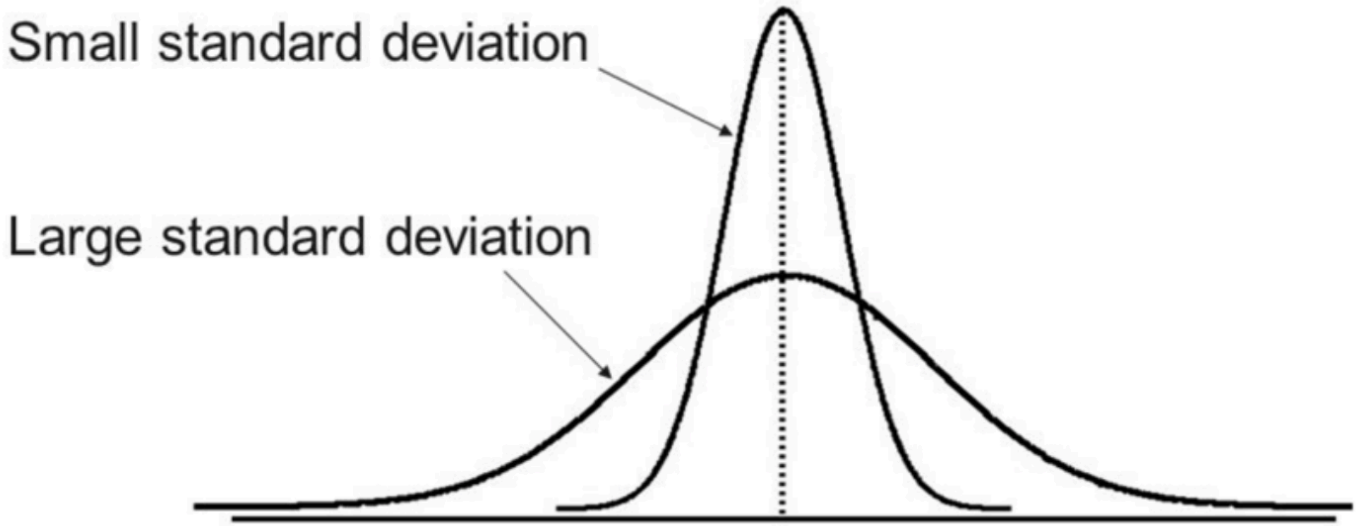
A sample case (cont.)

- Revenue per customer by cohort (another view)

Cohort	1st month	2nd month	3rd month	4th month	5th month
January	$5	$3	$2	$1	$0.5
February	$6	$4	$2	$1	
March	$7	$6	$5		
April	$8	$7			
May	$9				
Average	**$7**	**$5**	**$3**	**$1**	**$0.5**

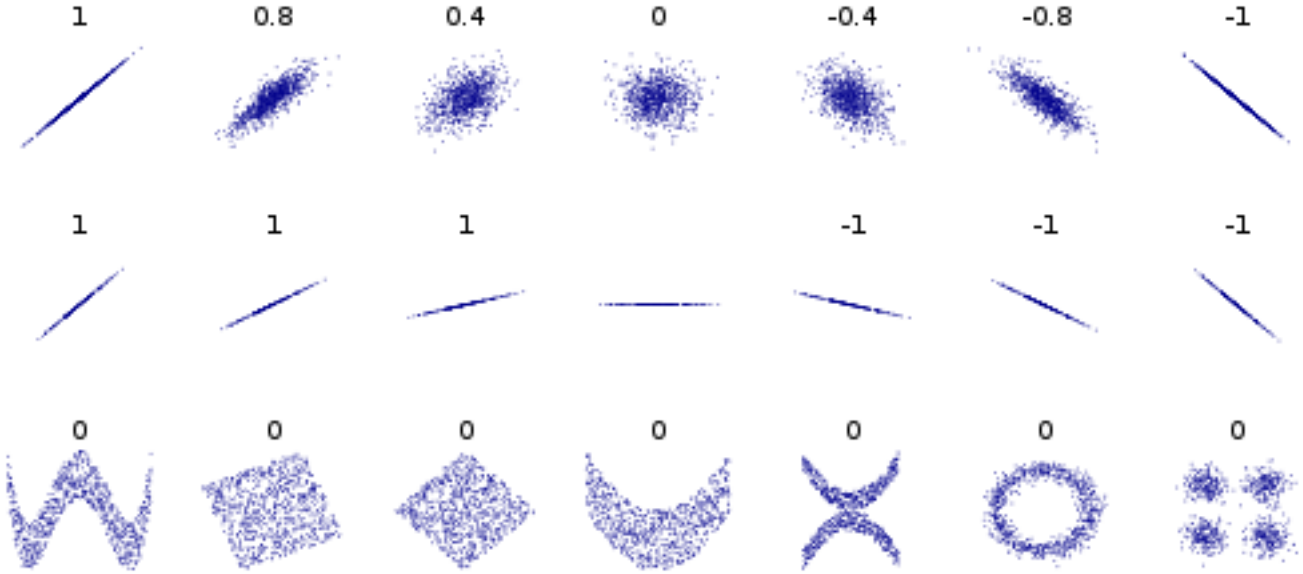
A few notes on statistics: variance and std. dev.

- Variance and standard deviation quantify the amount of variation or dispersion of a set of data values
  - Mean: $\bar{x}$
  - Variance: $\mathrm{Var}(X) = \sigma^2 = \frac{1}{n}\Sigma_{i=1}^{n}(\mathrm{x_i} - \bar{x})^2$
  - Standard deviation: σ



Covariance and Correlation

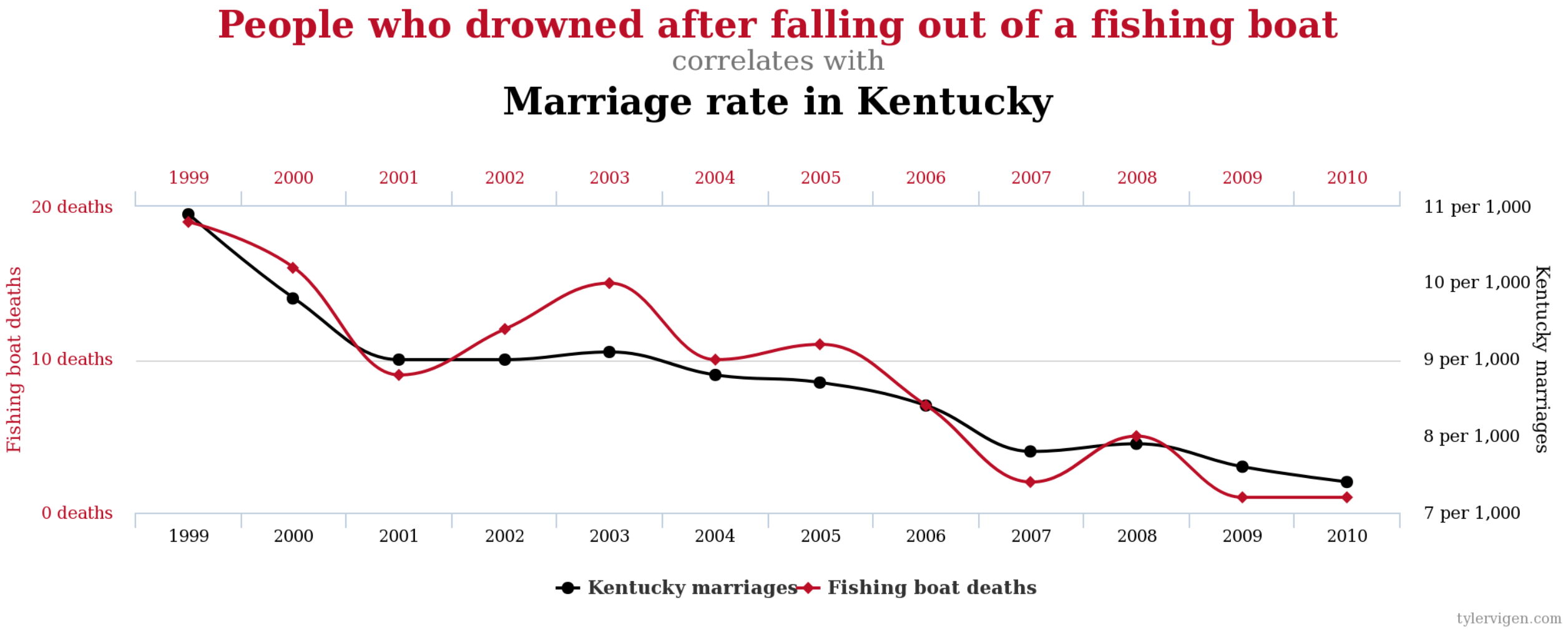
- Covariance and correlation measure of how much two variables change together
- When we have two variables x_i and y_i
  - $\mathrm{cov}(X,Y) = \frac{1}{n}\Sigma_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$
  - $corr(X,Y) = \frac{cov(X,Y)}{\sigma_x \sigma_y}$



Correlation v.s. Causation

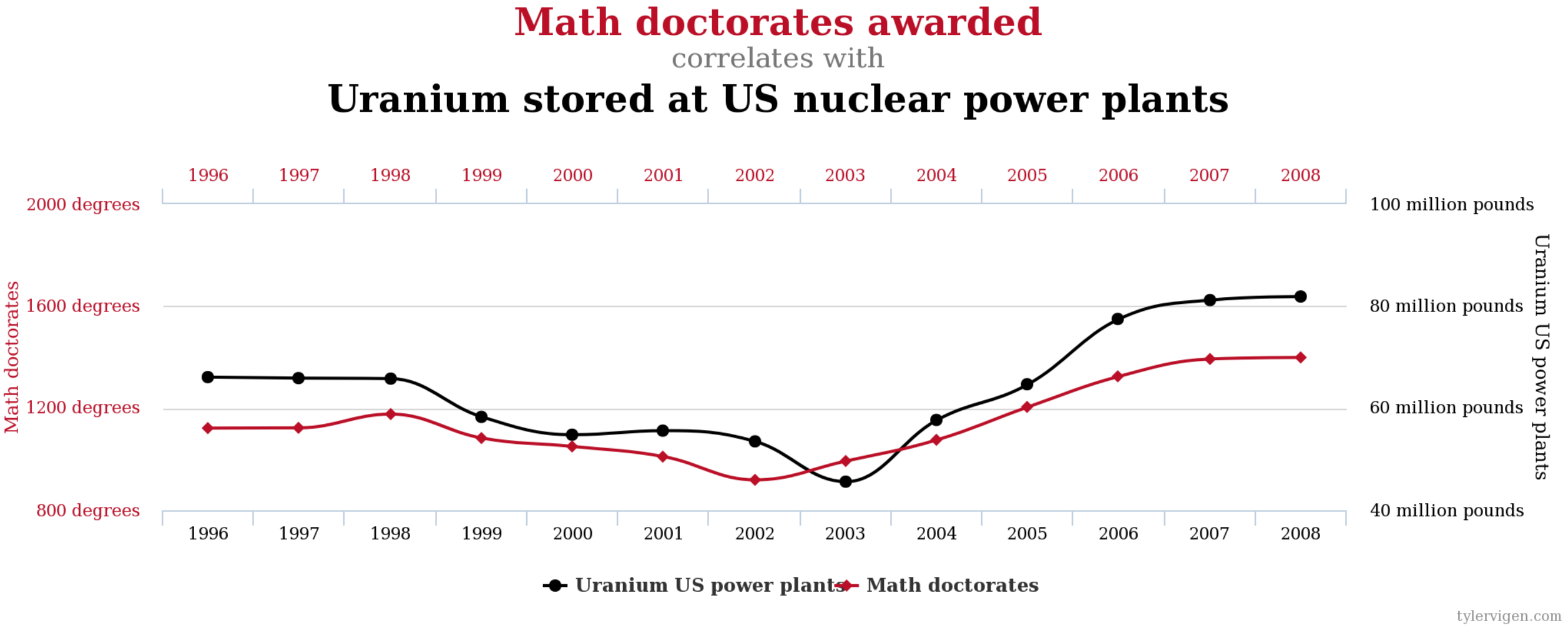
- Correlated: two variables that are related (but may be dependent on something else)
- Causal: an independent variable that directly impacts a dependent one

Spurious correlations



Source: www.tylervigen.com

Spurious correlations (cont.)



ANY QUESTIONS?